

Problem Definition

- Control Systems utilizes Quanser Qube-Servo 2 as a teaching platform.
- Drawbacks of the Qube-Servo 2
 - Costs \$5,000 per device
 - Requires outdated software
 - Not easily portable
 - Requires proprietary software
 - The Partnership Program only owns six
- The Electrical and Computer Engineering Program desires to increase the number of devices available.

Design Requirements

- Replicate all major functions of Qube-Servo 2
- Open-source
- Easy for students to carry around
- Able to be updated to work with future software versions
- Easy to take apart for maintenance and modification

Design Evaluation

Requirements	Testing Method	Results
Qube Function Replication	Complete Qube Servo 2 Tasks	Pending
Motor and Encoder Performance	Response Comparison	Pass
Data Transmission Speed	Response Timing	Pass
Open Source Documentation	Documentation Review	Pending
Cost Target	Hand Calculation	Pass
Self-Contained Storage	User Testing	Pass
Transport Protection	User Testing	Pass
Portability	User Testing	Pass
MATLAB/Simulink Compatibility	MATLAB/Simulink Connection	Pass
Software Updatability	Code Modification	Pending
Ease of Maintenance	User Testing	Pass
Third-Party Manufacturability	Procurement Validation	Pass

Design

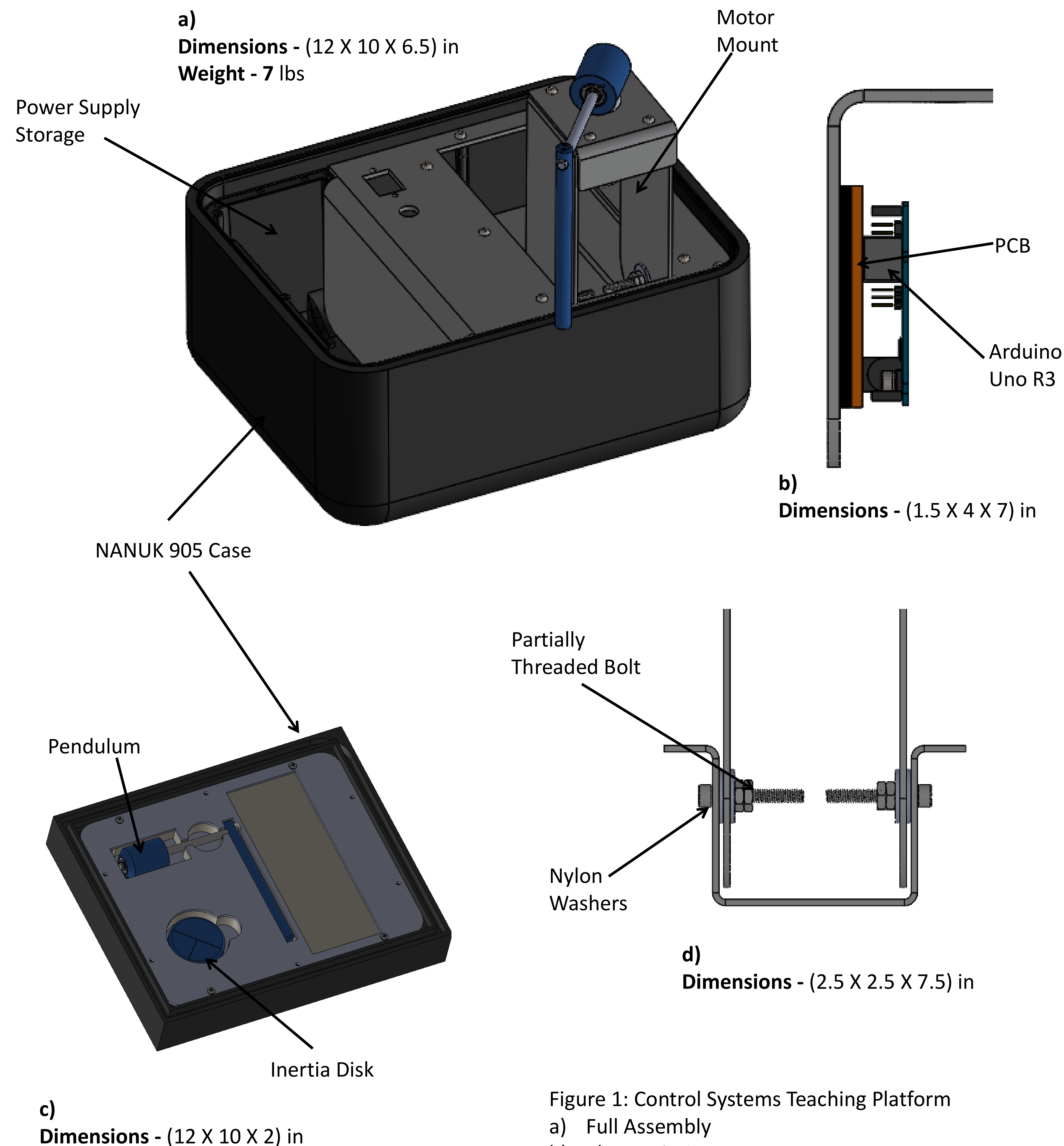


Figure 1: Control Systems Teaching Platform

- a) Full Assembly
- b) Electronic Compartment
- c) Lid Assembly
- d) Hinge Assembly

Electrical Design

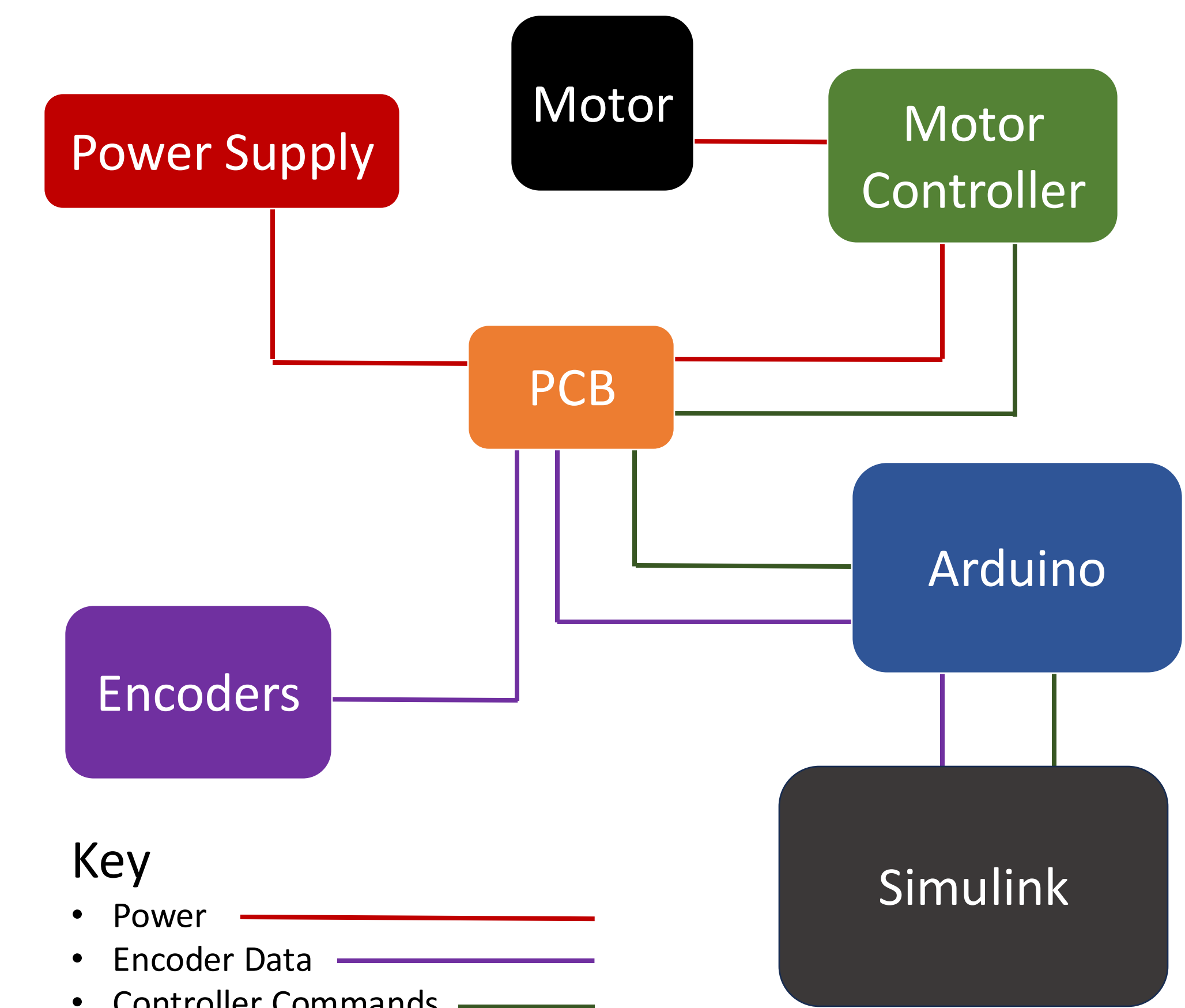


Figure 2: Block diagram of the electrical communication of the Control Systems Teaching Platform

Conclusions

- Development costs totaled to \$2,000
- Production costs estimated to be \$862.25 per unit
- System communicates data with speed and accuracy
- Storage and portability are improved in comparison to Qube-Servo 2
- Critical function and performance testing still in progress
- Final success depends on matching the functionality of the Qube-Servo

Next Steps

- Consolidate documentation, code, schematics, and instructions on GitHub.
- Catalog extra components remaining from research and development for reuse.
- Order parts for a small production run.
- Assemble units in batches using the provided instructions.
- Test each unit and prepare for classroom use.